Week : 12 (16-20 November 2026)

الأسبوع #: من - إلى

Subject/الموضوع: Physics **Unit Title عنوان الوحدة: Projectile Motion** **Grade الصف: 11**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: Report from the Courtyard – Results & Analysis	I can process my courtyard data into clear results tables, calculate average range and time of flight, determine velocity components and theoretical range, and graph range against launch angle (DOK 2–3).	6.2 Support planning with templates, strategy prompts, and modeled thinking.	Week 11 courtyard data tables, photos and video; scientific report template on Toddle; calculator, Excel/Google Sheets, GRASP formula sheet.
Lesson 2: Report from the Courtyard – Evaluation & Conclusion	I can compare theoretical and experimental range using percentage difference, evaluate sources of error such as air resistance and measurement uncertainty, and write an evidence-based conclusion recommending an optimal launch angle (DOK 3–4).	5.2 Use multiple tools for composition (spellcheckers, manipulatives, templates). 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	Scientific report template and GRASP rubric on Toddle; graphs from Lesson 1; Physics notebook.
Lesson 3: GRASP Presentations (Part 1)	I can present my findings to the ADNOC Sports Academy panel, explaining my method, data, calculations and recommended launch angle, and answer questions using physics evidence (DOK 3–4).	5.1 Provide multiple communication media (text, speech, visuals, models). 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement.	GRASP PowerPoint, GRASP rubric on Toddle, projector/screen, peer feedback form.
Lesson 4: GRASP Presentations (Part 2) & Reflection	I can justify my recommendation with evidence during my presentation, give constructive peer feedback, and reflect on my learning against the GRASP rubric (DOK 3–4).	5.1 Provide multiple communication media (text, speech, visuals, models). 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	GRASP PowerPoint, GRASP rubric and submission instructions on Toddle, peer feedback and self-reflection forms.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Lessons 1–2: students write their scientific report using the data collected in the courtyard in Week 11. Bring calculator, data tables and Physics notebook. Lessons 3–4: GRASP presentations; the presentation order is shared on Toddle and the audience completes peer feedback. Final report and PowerPoint are submitted on Toddle.		

Week #12 الأسبوع#

Technology Integration دمج التكنولوجيا	Excel/Google Sheets data processing and graphing; experiment photos/video as evidence; PowerPoint presentation; Toddle submission.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: trigonometry, averages, graphing and percentage difference. PE: ball sports and launch angles. ICT: video analysis.
External Assessment Integration دمج التقييمات الخارجية	NGSS scientific practices: planning investigations, analysing data and communicating evidence-based explanations.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking: compare theoretical and experimental range. Research: collect reliable measurements. Communication: explain conclusions. Collaboration and wellbeing: shared roles and safe practical work.
Standards المعايير	NGSS HS-PS2-1 — Analyze data to support the claim that Newton's second law describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can a UAE sports engineer determine the launch angle that gives a ball the greatest range, using real measurements and physics?
Success Criteria معايير النجاح **(Measurable)**	MUST (DOK 1–2): Measure and record launch angle, range and flight time, with labelled evidence and units. SHOULD (DOK 3): Calculate velocity components, maximum height and range; graph range versus angle; recommend an angle using data. COULD (DOK 4): Critically evaluate discrepancies, air resistance and measurement uncertainty; propose realistic improvements.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء — Global Understanding التفاهم العالمي — Arabic Language اللغة العربية
	Area: Innovation and Infrastructure / UAE sports excellence
	Implementation: Students act as sports engineers for ADNOC Sports Academy, applying projectile-motion data to UAE sporting contexts such as Zayed Sports City and Khalifa Stadium. They develop evidence-based recommendations that connect physics, innovation and athletic performance.